Loan Default Prediction System

A Comparative Study of Classification Algorithms

Mini Project | Supervised Learning Module

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Abstract

Banks approve loans every day without knowing for certain who will repay and who will default. This project builds a small predictive system that estimates the risk of default from three customer attributes employment status, bank balance, and annual salary using a real bank dataset of 10,000 customers. Four classifiers (Logistic Regression, Random Forest, Support Vector Machine, and XGBoost) are trained, tuned for class imbalance, and compared on accuracy, precision, recall, F1-score, and ROC-AUC. The best-performing model is selected and its predictions are explained using feature importance analysis.

1 Problem Statement

A bank wants to flag customers who are likely to default on a loan *before* approving it, so it can price the loan appropriately or decline high-risk applicants. The task is framed as a binary classification problem: given a customer's employment status, bank balance, and annual salary, predict whether they will default (`1`) or not (`0`).

Key Idea

This is a classic **imbalanced classification** problem defaults are rare events. A model that always predicts "no default" would already be right most of the time, so accuracy alone is a misleading metric here.

2 Dataset Description

The dataset (`Default_Fin.csv`) contains 10,000 customer records with the following columns:

Column	Description
Employed	1 if the customer is employed, 0 otherwise
Bank Balance	Current bank balance (currency units)
Annual Salary	Annual income (currency units)
Defaulted?	Target variable 1 if the customer defaulted, 0 otherwise

The `Index` column was dropped as it carries no predictive information. The dataset has no missing values.

2.1 Class Balance

Only 3.33% of customers (333 out of 10,000) defaulted, confirming this is a highly imbalanced dataset.

Watch Out

With only 3.33% positive cases, a naive classifier that predicts "no default" every time would score 96.67% accuracy while being completely useless. All models below use class weighting to counter this.

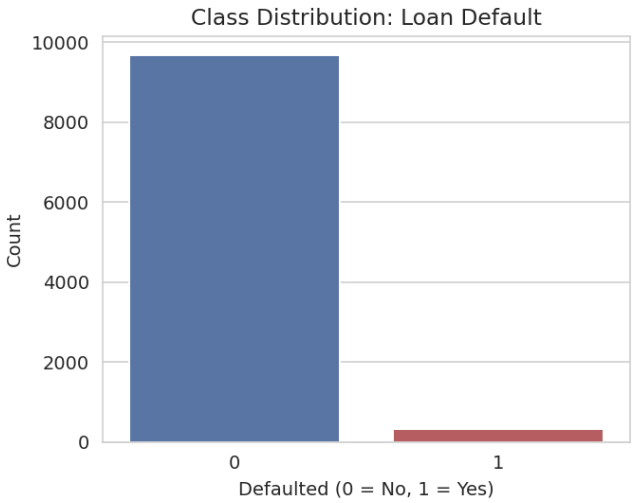


Figure 1: Class distribution of the target variable. Defaults make up only a small fraction of the dataset.

2.2 Feature Correlations

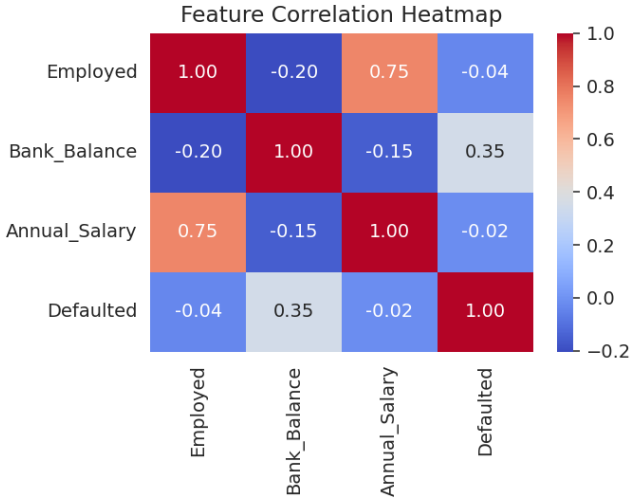


Figure 2: Correlation heatmap of the numeric features and the target.

Intuition

Bank balance shows the strongest correlation with default – customers with lower balances are more likely to default, which matches real-world lending intuition. Salary has a weaker positive relationship, and employment status barely correlates at all once balance is accounted for.

3 Methodology

3.1 Preprocessing

- Dropped the non-predictive `Index` column.

- Split the data into 80% training and 20% test sets, using stratified sampling to preserve the 3.33% default rate in both sets.
- Standardized features (zero mean, unit variance) for Logistic Regression and SVM, since both are sensitive to feature scale. Random Forest and XGBoost were trained on the raw, unscaled features since tree-based models are scale-invariant.

3.2 Handling Class Imbalance

Each model was configured to give more weight to the minority (default) class during training:

- Logistic Regression, Random Forest, and SVM used `class_weight=balanced`.
- XGBoost used `scale_pos_weight`, set to the ratio of negative to positive samples in the training set (≈ 29).

3.3 Models Compared

Model	Key hyperparameters
Logistic Regression	`max_iter=1000`, balanced class weights
Random Forest	`n_estimators=300`, balanced class weights
SVM (RBF kernel)	`probability=True`, balanced class weights
XGBoost	`n_estimators=300`, `max_depth=4`, `learning_rate=0.05`

3.4 Evaluation Metrics

Because the classes are imbalanced, five metrics were tracked rather than accuracy alone: **accuracy**, **precision**, **recall**, **F1-score**, and **ROC-AUC**. Recall is especially important here – missing an actual defaulter (a false negative) is typically far more costly to a bank than flagging a safe customer as risky (a false positive).

4 Results

4.1 Performance Comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.855	0.173	**0.881**	0.289	**0.949**
XGBoost	0.889	0.202	0.791	0.322	0.937
SVM (RBF)	0.841	0.162	0.896	0.274	0.915
Random Forest	**0.970**	**0.594**	0.284	**0.384**	0.889

Table 1: Test-set performance of all four classifiers, sorted by ROC-AUC.

Worked Example

Random Forest has the highest *accuracy* (97%) but the *lowest recall* (28%) among the four models – it plays it safe and rarely predicts a default, so it misses most of the actual defaulters. This is exactly the trap accuracy sets for imbalanced data: a high accuracy score can hide a model that fails at the one thing it was built to do.

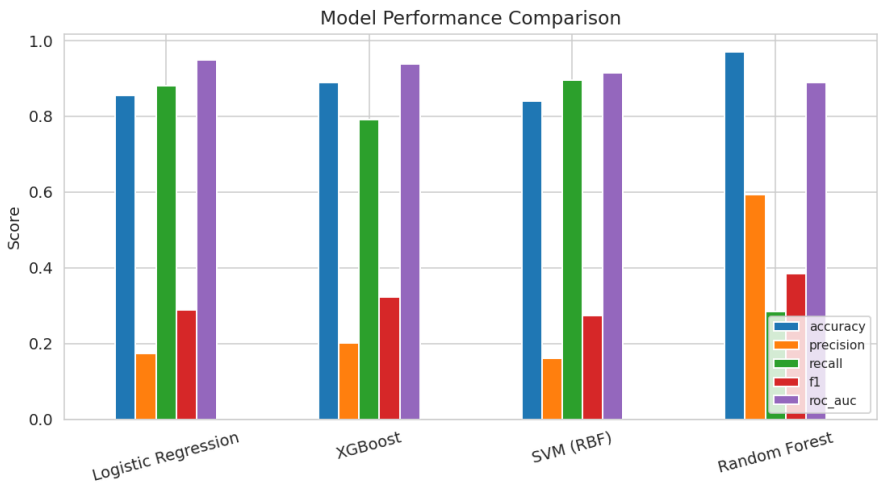


Figure 3: Side-by-side comparison of all five metrics across the four models.

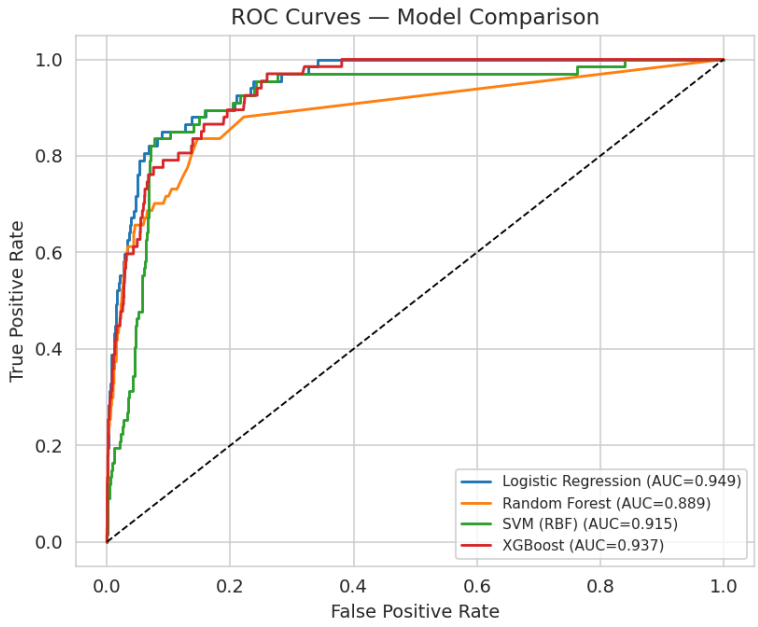


Figure 4: ROC curves for all four models. Logistic Regression has the largest area under the curve.

4.2 Selecting the Best Model

Ranking by ROC-AUC the metric least distorted by class imbalance **Logistic Regression comes out on top** (AUC = 0.949), followed closely by XGBoost (0.937). Logistic Regression also achieves the second-highest recall (88%), meaning it correctly flags the large majority of customers who go on to default. Its precision is lower, so it does raise more false alarms, but for a default-screening system that trade-off is usually acceptable a missed defaulter is more expensive than a few extra manual reviews.

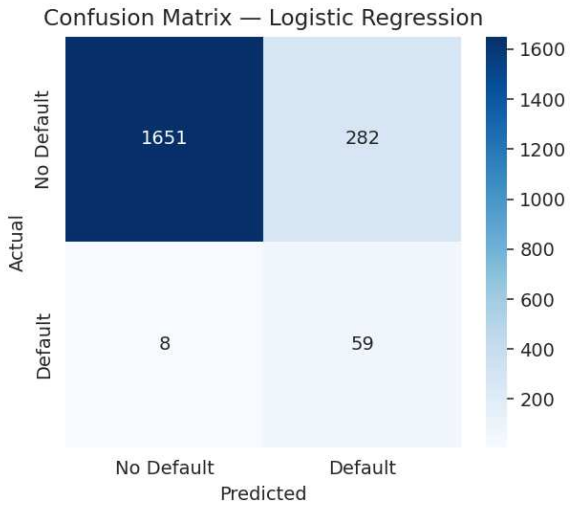


Figure 5: Confusion matrix for the selected best model, Logistic Regression.

5 Feature Importance

Logistic Regression does not produce a tree-style importance score, so feature importance was examined using the Random Forest and XGBoost models, both of which agree on the ranking.

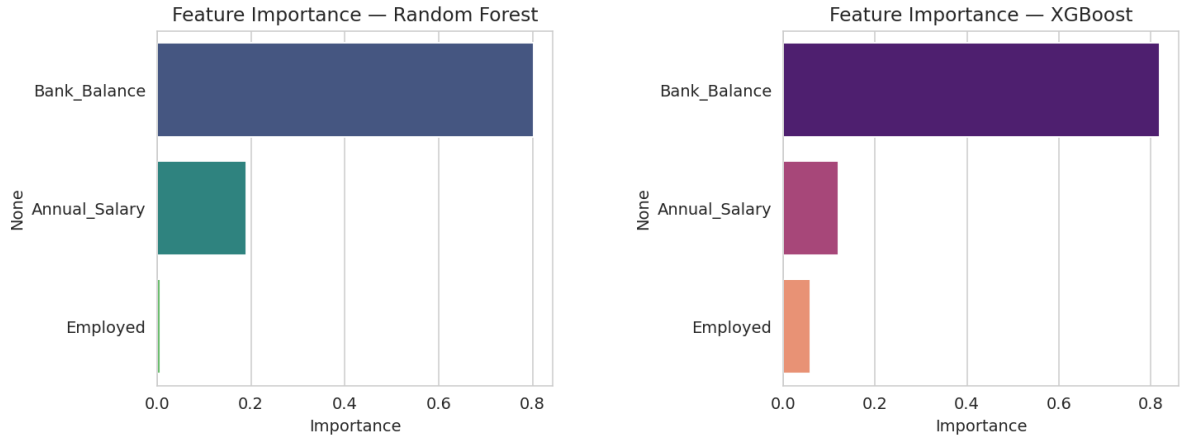


Figure 6: Feature importance from Random Forest (left) and XGBoost (right). Both models agree that bank balance dominates.

Key Idea

Bank balance is, by a wide margin, the strongest predictor of default (roughly 80% of Random Forest's total importance), followed by annual salary. Employment status contributes very little once balance and salary are known – a customer's cash reserves matter far more than whether they currently hold a job.

6 Conclusion

This project compared four classifiers for predicting loan default on an imbalanced banking dataset. The key findings are:

1. Accuracy is a misleading metric on this dataset Random Forest scores highest on accuracy (97%) but performs worst on recall (28%), meaning it misses most real defaulters.
2. **Logistic Regression** is the recommended model, combining the best ROC-AUC (0.949) with strong recall (88%), making it the most reliable at catching risky customers.
3. **Bank balance** is by far the most important feature driving default risk, followed by annual salary, with employment status playing a minor role.
4. In a production setting, the choice between Logistic Regression (higher recall, more false alarms) and XGBoost (a more balanced precision/recall trade-off) should be guided by the bank's actual cost of a missed default versus a false alarm.

Tools Used

Python, scikit-learn, XGBoost, pandas, NumPy, Matplotlib, Seaborn.